Development summary: spherical Regge scissors congruence

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Discussions and model prompting: Alexander Kolpakov (UATX). This retrospective account describes the focused conditional manuscript and its mathematical documentation. It is not a transcript or a reconstruction of model turns. No original conversation export or authenticated model identifier was supplied with the focused release.

Schematic research brief

The target is to show that a nondegenerate spherical tetrahedron and its elementary Regge mate admit finite geodesic dissections with pieces paired by spherical isometries. A schematic brief is to identify the relevant scissors-class defect, separate the finite algebra from the deeper comparison theorems, and make every required input explicit. This is an editorial description of the task, not a quotation of an original prompt.

The released result is conditional: the conclusion follows from the numbered package E1–E8. The relative motivic and comparison bridge E2–E5 has not been constructed. The unconditional spherical Regge scissors theorem is therefore not established by this contribution.

Mathematical route

1. **Encode the elementary Regge operation.** The four moving lengths and angles transform by $R = \frac{1}{2}J - I$, where J is the all-ones matrix. Prove the matrix involution and orthogonality. Phase variables turn the corresponding angle transformation into $a_i \mapsto \tau/a_i$ with $\tau^2 = \prod_i a_i$.

2. **Check the orientation-cover algebra.** Expand the Gram determinant, prove its relevant coefficient polynomials invariant under the phase substitution, and verify the two rational chart inverses and their cover on the stated open locus. Check the fixed-phase polynomial and its deck transformation.

3. **Expose the relative mathematical input.** Specify relative framed quadric coefficients, a strict three-weight setting, and the primitive-kernel comparison with $H^1(U, \mathbb{Q}(2))$. The proposed vanishing argument depends on the exact relative package; descriptions of its intended meaning do not supply that construction.

4. **Derive the finite channel cancellation.** The four moving channels cancel through $H^t H = 4I$; the two fixed channels agree. Given the six-channel coproduct formula and descent in E4, the universal defect is primitive. E3 then forces it to vanish.

5. **Follow specialization to the spherical class.** Use the supplied commuting square, endpoint comparison, field-level injections and presentation comparison to kill the reduced scissors defect. Suspension, equal volume, area classification and cancellation then give a finite equidecomposition under the remaining inputs.

6. **Check the implication and its boundary.** Match every labeled manuscript item and every external-contract field to the formal declarations. Keep the unconstructed relative bridge explicit in the theorem statement and in the public account.

Formal certificate and remaining work

The Lean project checks the matrix, phase, determinant, chart, deck and tensor algebra, followed by the kernel and scissors-group deductions. Its main theorem takes a supplied `ProofContract` and returns a typed finite-list equidecomposition certificate. Its abstract geometric predicates and carriers are interpretation obligations; they do not build actual spherical tetrahedra, motives or scissors groups from foundations.

The unresolved E2–E5 work includes constructing the relative face objects, establishing the strict three-weight/primitive comparison, proving the relative coproduct and Artin descent, and matching the specialized endpoint with the field-level coefficient with the required signs and frames. Published field-level results do not by themselves supply this exact relative package. The Lean axiom audit checks the conditional implication, not the existence of a geometric instance of its hypotheses.

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